Uthuralakshmi M

AI & Generative AI Engineer

Chennai, Tamil Nadu | +91 8925132163 | Email | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Aspiring AI and Generative AI Engineer with hands-on experience developing Machine Learning models, RAG-based document assistants, and AI-powered agents. Skilled in Python, data preprocessing, Scikit-learn, LLM integration, embeddings, vector databases, and tool-calling workflows. Experienced in building practical AI applications combining language models with Python tools, document retrieval, structured outputs, and human approval workflows.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Data Analysis: NumPy, Pandas, Exploratory Data Analysis, Data Cleaning, Matplotlib, Seaborn

Machine Learning: Scikit-learn, Linear Regression, Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, Feature Engineering, Model Evaluation, Cross-Validation, GridSearchCV

Generative AI: LLMs, Prompt Engineering, Gemini API, LLM Integration, Embeddings, Retrieval-Augmented Generation, Semantic Search

AI Agents: Tool Calling, Function Calling, Agent Workflows, Human-in-the-Loop

Vector Databases and Reliability: ChromaDB, Vector Search, JSON Schema Validation, LLM Guardrails, PII Detection

Developer Tools: Git, GitHub, Google Colab, Jupyter Notebook

PROJECTS

Advanced RAG Document Assistant GitHub

- Built a PDF question-answering application using Retrieval-Augmented Generation, Gemini API, and ChromaDB.
- Processed a 352-page document into 1,504 chunks, generated 384-dimensional embeddings, and implemented semantic retrieval with page citations and evidence checks.

AI Customer Support Agent – SupportPilot AI GitHub

- Developed an AI-powered support agent for order lookup, return requests, and customer queries.
- Implemented return workflows with human approval and escalation for unsupported questions.

Data Analyst AI Agent GitHub

- Built a Gemini-powered data analysis agent using Python, Pandas, and tool calling.
- Created analytical tools for top products, monthly sales, category average order value, and trend detection.

Hotel Booking Cancellation Prediction GitHub

- Developed a booking cancellation prediction pipeline using Python, Pandas, and Scikit-learn.
- Applied preprocessing, feature engineering, classification models, ensemble learning, and GridSearchCV.
- Integrated GPT-based prediction explanations with JSON Schema validation and PII guardrails.

EDUCATION

Bachelor of Commerce (B.Com) – Corporate Secretaryship 2022 – 2025
Sri Paramakalyani College, Alwarkurichi

CERTIFICATIONS

Applied Artificial Intelligence and Machine Learning
Masai School / Vishleshan i-HUB IIT Patna